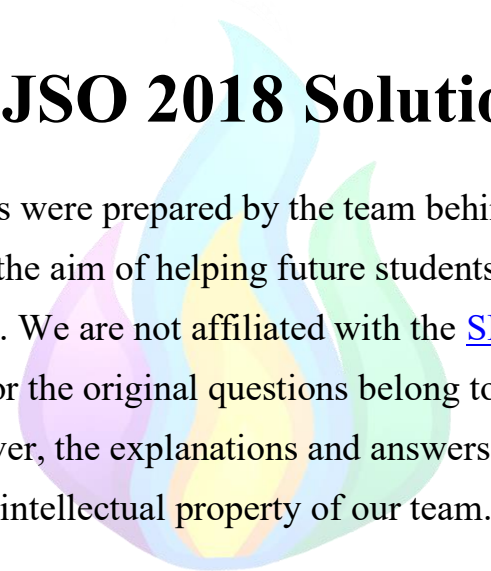




SLJSO 2018 Solutions

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Question No.	Field	References
<u>1</u>	Cell Biology	<u>Cellular Respiration</u>
<u>2</u>	Cell Biology	<u>Prokaryotic Cells</u>
<u>3</u>	Laboratory Guide	Benedict's Test
<u>4</u>	Ecology	<u>Symbiosis</u>
<u>5</u>	Botany	<u>Photosynthesis</u>
<u>6</u>	Human Biology	<u>Circulatory System</u>
<u>7</u>	Human Biology	<u>Homeostasis</u>
<u>8</u>	Genetics	<u>Alleles</u>
<u>9</u>	Genetics	<u>Punnett Squares</u>
<u>10</u>	Microbiology	<u>Microbes</u>
<u>11</u>	Cell Biology	<u>Cellular Respiration</u>
<u>12</u>	Cell Biology	<u>Passive Transport</u>
<u>13</u>	Human Biology	<u>Endocrine System</u>
<u>14</u>	Evolutionary Development	<u>Animal Classification</u>
<u>15</u>	Evolutionary Development	<u>Pathogens</u>
<u>16</u>	Chemical Structures	<u>Electronic Configurations</u>
<u>17</u>	Chemical Compounds	
<u>18</u>		
<u>19</u>	Thermodynamics	<u>Ideal Gas Law</u>
<u>20</u>	Laboratory Guide	Mixtures
<u>21</u>		
<u>22</u>	Stoichiometry	<u>Concentration</u>
<u>23</u>	Stoichiometry	<u>Concentration</u>
<u>24</u>	Stoichiometry	<u>Concentration</u>
<u>25</u>	Thermochemistry	<u>Lattice Energy</u>
<u>26</u>	Chemical Structures	<u>Isotopes</u>
<u>27</u>	Stoichiometry	<u>Mole</u>

<u>28</u>	Stoichiometry	<u>Mole</u>
<u>29</u>	Stoichiometry	<u>Limiting Reactant</u>
<u>30</u>	Cellular Biology	<u>Sugars</u>
<u>31</u>	Mechanics	<u>Work</u>
<u>32</u>	Waves	<u>Wave Equation</u>
<u>33</u>	Electromagnetism	<u>Series and Parallel Circuits</u>
<u>34</u>	Units	<u>Dimensions and Analysis</u>
<u>35</u>	Thermodynamics	<u>Ideal Gas Law</u>
<u>36</u>	Thermodynamics	<u>Specific Heat Capacity</u>
<u>37</u>	DELETED QUESTION	
<u>38</u>	Mechanics	<u>Motion Graphs</u>
<u>39</u>	Mechanics	<u>Pressure</u>
<u>40</u>	Thermodynamics	<u>Specific Heat Capacity</u>
<u>41</u>	Light	<u>Mirrors</u>
<u>42</u>	Mechanics	<u>Power</u>
<u>43</u>	Light	<u>Lenses</u>
<u>44</u>	Mechanics	<u>Forces</u>
<u>45</u>	Mechanics	<u>Pressure</u>
<u>46</u>	Electromagnetism	<u>Ohm's Law</u>
<u>47</u>	Mechanics	<u>Buoyancy</u>
<u>48</u>	Mechanics	<u>Energy Conservation</u>
<u>49</u>	DELETED QUESTION	
<u>50</u>	Algebra	

Question 1

This problem is tackling metabolic processes, specifically cellular respiration. Option (A) is describing glycolysis, which has a net gain of 2 ATP molecules. Option (B) is describing cellular respiration as a whole, which produces 32 - 38 ATP (depending on the cell). Option (C) is describing lactic acid fermentation, and option (D) is describing alcoholic fermentation, both of which produce only 2 molecules of ATP per molecule of glucose. By comparing the numbers, we find that option (B) $\text{Glucose} \rightarrow \text{CO}_2$ is the correct answer.

Correct Answer: B

Question 2

This problem is focusing on prokaryotic cell structure and function. Option (A) says that prokaryotic cells contain membrane-bound organelles, which is incorrect. Options (B) and (C) both describe things that are found in both eukaryotic and prokaryotic cells. Option (D) is the correct answer because nitrogen-fixing bacteria are the only organisms with the ability to fix atmospheric nitrogen, and all bacteria are prokaryotic.

Correct Answer: D

Question 3

Benedict's solution is used to detect the presence of reducing sugars by turning red, like glucose, fructose, galactose, lactose, and maltose. Option I is a glucose solution, which is a reducing sugar, so it will turn red. In Option II, HCl hydrolyzes (breaks apart with water) sucrose into glucose and fructose—both of which are reducing sugars, turning red. In Option III, starch is broken down by amylase into maltose and glucose—both of which are reducing sugars, turning red. Therefore, Option (D) is the correct answer, as all of the solutions form a red precipitate.

Correct Answer: D

Question 4

Option (A) is considered symbiosis because the fungi helps to decompose and transform nutrients to make them available for the plants. The plants give back to the fungi by providing them with sugars from photosynthesis. Option (B) is considered symbiotic—specifically parasitism—the cuscuta plant is taking nutrients from the plant without giving anything back. However, symbiosis can be either beneficial, harmful, or neutral, as long as it involves a close relationship between two or more organisms. Option (C) is considered symbiotic, as the bacteria fix nitrogen into usable nitrogen for the plant, whilst the plant provides bacteria with sugars produced by photosynthesis. Option (D) is not considered symbiotic, as the lichen is interacting with the fungi and the moss ON the bark, NOT the bark itself. This indicates the option (D) is the correct answer.

Correct Answer: D

Question 5

Photosynthesis is an energy-absorbing process (it takes energy from photons and splits them to get electrons). Option (A) is false for that reason. Option (B) is true—chlorophyll is best at absorbing red and blue light, as seen in an absorption spectra graph. Option (C) is true—there are other pigments like carotenoids and xanthophylls that can be a part of photosynthesis. Option (D) is true; oxygen in photosynthesis comes from the splitting (photolysis) of water. Therefore, Option (A) is the correct answer.

Correct Answer: A

Question 6

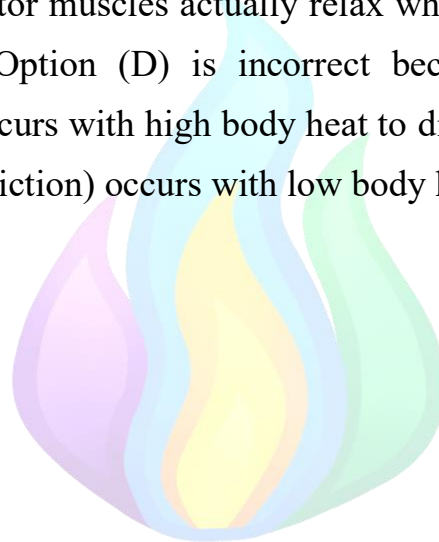
Option (A), Vitamin A, is essential when it comes to vision, skin, and immune health. Option (B), Vitamin B, is essential for energy metabolism and red blood cell formation. Option (C), Vitamin C, is essential for collagen synthesis and antioxidation. Therefore, Option (D), Vitamin K, is essential for the synthesis of clotting factors in the liver (eg: prothrombin).

Correct Answer: D

Question 7

Option (A) is incorrect because evaporation of water from the skin (sweat) is a cooling mechanism that occurs due to an increase in body temperature, not a decrease. Option (B) is correct. Metabolism expends energy, releasing heat. Hormones like thyroxine and epinephrine regulate metabolism, which in turn regulates body temperature. If body temperature is high, these hormones will lower metabolic rate to lower body temperature. Option (C) is incorrect because hair erector muscles in the skin contract when body temperature is low, known as “goosebumps.” Hair erector muscles actually relax when body temperature is high to easily release heat. Option (D) is incorrect because blood vessel dilation (vasodilation) actually occurs with high body heat to dissipate heat. Constriction of blood vessels (vasoconstriction) occurs with low body heat to retain heat.

Correct Answer: B



Question 8

This question is focusing on genetic variation. Option (A), independent segregation, does in fact increase genetic variation, as alleles can segregate independently of one another, increasing randomness. Option (B), linkage, is the correct option, as linked genes are inherited together—decreasing the amount of variation available. Option (C), crossing over, increases genetic variability by allowing for swapping of alleles. Option (D), incomplete dominance, increases incomplete dominance as it leads to a wider range of phenotypes.

Correct Answer: B

Question 9

This is a genetic cross question. Right away, we know that the plant with white flower and yellow seeds must have the genotype $rrgg$, as white flowers and yellow seeds are recessive, so in order to exhibit those traits, the individual must be homozygous recessive. We can also generate all the possible combinations of the offspring's genotypes: red flowers with green seeds could be one of the following: $RRGG$, $RrGg$, $RRGg$, or $RrGG$. The white flowers with green seeds could be $rrGG$ or $rrGg$. Testing each one in a genetic cross led to the correct answer being option (B), $RrGG$ as when it crosses with $rrgg$, it produces $RrGg$ (making red flowers with green seeds) and $rrGg$ (making white flowers with green seeds).

Correct Answer: B

Question 10

Option (A) is true; the intestine is home to trillions of microorganisms aiding in enzyme production and digestion. Option (B) is true because pathogens can release enzymes to aid in invasion and break down the host cell. Option (C) is false, making it correct, because endotoxins are heat stable. Exotoxins are the unstable ones. Option (D) is true, as the skin is the first protective layer against pathogens and other dangers, protecting the body from physical and chemical threats.

Correct Answer: C

Question 11

Option (A) is true; ATP is a nucleotide (it consists of an adenine base attached to a ribose sugar with 3 phosphate groups). Option (B) is also true; ATP is produced during photosynthesis. Option (C) is false; the most ATP per molecule of glucose is actually produced during the ETC. Option (D) is true—ATP is used in muscle contraction. Therefore, option (C) is the correct answer.

Correct Answer: C

Question 12

In this question, we are looking for a passive process. Passive processes don't require energy input—they are spontaneous. Option (A) is an active process, using energy to pull water up. Option (B) is also an active process; the movement of sucrose from mesophyll cells into sieve tubes requires energy. Option (C) is a passive process—sucrose molecules move passively from sieve tube to sieve tube based on a pressure gradient. Option (D) is an active process, as ion movement in the proximal convoluted tubule is an active process.

Correct Answer: C

Question 13

Option (A) is incorrect because the skin is actually the largest organ of the body. Option (B) lies in the right upper region of the abdomen, so it is incorrect. Option (C) is incorrect because bone marrow synthesizes hemoglobin, not the liver. Option (D) is correct because it produces bile, which plays a role in digestion of food

Correct Answer: D

Question 14

Option (A) is true because skeletons assist in locomotion. Option (B) is true because annelids do in fact have a hydrostatic skeleton. Option (C) is also true, as exoskeletons are typically confined to their exoskeleton and grow into them. Option (D) is false, making it correct. Endoskeletons (not exoskeletons) are found in some invertebrates as well, like echinoderms and cephalopods.

Correct Answer: D

Question 15

Option (A) is false, making it correct. Most plant viruses contain RNA. Option (B) is true; viruses can have either DNA or RNA. Option (C) is true; viruses can be used as vectors to deliver genetic material, which is helpful in recombinant technology. Option (D) is true; all viruses are obligate parasites, as they require a host to replicate and cannot survive outside of a cell.

Correct Answer: A

Question 16

N^{3-} (10 e^-), Na^+ (10 e^-), and Ne (10 e^-) are isoelectronic. Oxygen (O) has 8 e^- , so it is not isoelectronic.

Correct Answer: D

Question 17

- (A) True—Concentrated HCl is denser than water.
- (B) False—Laboratory HCl is an aqueous solution of HCl gas, not pure HCl.
- (C) True—HCl reacts with NH_3 to form NH_4Cl solid. However this solid appears as a white smoke or fumes.
- (D) True—HCl is highly corrosive.

Correct Answer: B

Question 18

The correct order of abundance in the Earth's crust is:

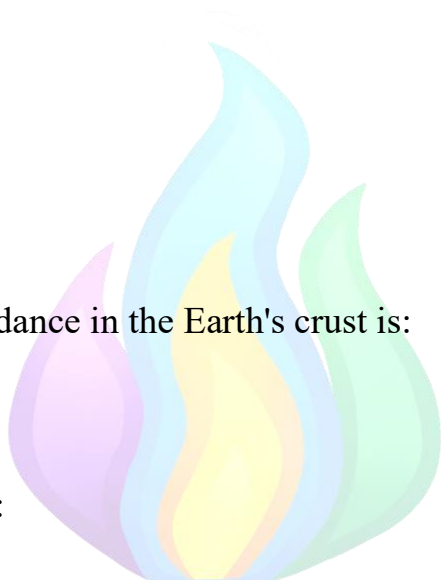
$\text{O} > \text{Si} > \text{Al} > \text{Fe}$

Approximate percentages:

Oxygen (O) – 46.6% Silicon (Si) – 27.7% Aluminum (Al) – 8.1% Iron (Fe) – 5.0%

Oxygen is the most abundant element in the Earth's crust, followed by silicon, which forms silicate minerals. Aluminum comes next, mainly found in feldspar and clay minerals, and iron is less abundant but still significant in minerals like hematite and magnetite.

Correct Answer: B



Question 19

Using the ideal gas law and Boyle's Law:

$$P_1 V_1 = P_2 V_2$$

Given:

- Initial volume = 3 L
- Initial pressure = 5 atm

$$\text{Final volume} = 3 \text{ L} + 2.5 \text{ L} = 5.5 \text{ L}$$

$$P_2 = \frac{P_1 V_1}{V_2} = \frac{5 \times 3}{5.5} = \frac{15}{5.5} \approx 2.73 \text{ atm}$$

When we consider the volume of the tap, which is not given, the actual total volume will be higher. This means the actual pressure will be lower. Since the closest (but lower) option to 2.73 atm is 2.5 atm, the correct choice is Option D.

Correct Answer: D

Question 20

22-carat gold is an alloy, meaning it is a homogeneous mixture of gold with other metals like copper or silver to improve strength.

- (A) Incorrect—It is not heterogeneous; alloys are uniform throughout.
- (B) Incorrect—It is not a pure compound but a mixture.
- (C) Correct—It is a homogeneous mixture.
- (D) Incorrect—Gold is highly resistant to oxidation.

Correct Answer: C

Question 21

The best solvent with minimal health risks for removing jack latex is acetone.

- (A) Hexane—Can be toxic with prolonged exposure.
- (B) Benzene—Highly carcinogenic and harmful.
- (C) Acetone—Safest choice, commonly used in nail polish remover, and effectively dissolves latex.
- (D) Liquid ammonia—harsh and can cause skin irritation.

Correct Answer: C

Question 22

To prepare an 8% w/w (weight/weight) aqueous solution, the total weight of the solution must be 100 g, consisting of 8 g of glucose and 92 g of water.

- (A) Incorrect—This results in 7.4% w/w instead of 8%.
- (B) Incorrect—uses volume (mL) instead of weight (g), which is not precise.
- (C) Incorrect—Contains too much water, making the solution highly diluted.
- (D) Correct—Dissolve 8 g of glucose in water and adjust until the total weight reaches 100 g.

Correct Answer: D

Question 23

To calculate the weight percentage (w/w) of glucose in the solution, use the formula:

$$\% \frac{w}{w} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 100$$

$$\% \frac{w}{w} = \frac{18}{268} \times 100 = 6.7\%$$

Correct Answer: D

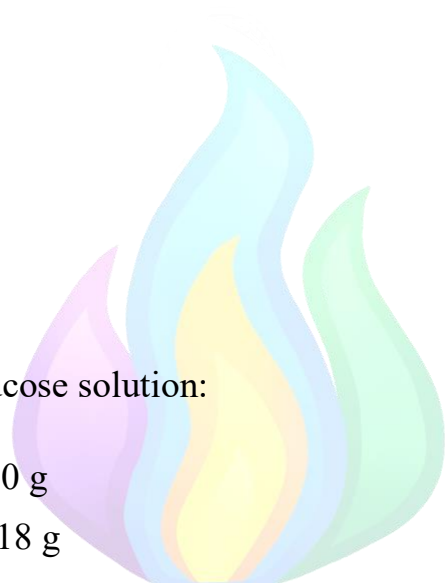
Question 24

Assuming a 10% w/w glucose solution:

- Mass of water = 250 g
- Mass of glucose = 18 g
- Molar mass of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) = 180 g/mol
- Moles of glucose = $18 \text{ g} / 180 \text{ g/mol} \approx 0.1$

$$\text{Molarity} = 0.1 \text{ mol} / 0.250 \text{ L} = 0.4 \text{ M}$$

Correct Answer: A



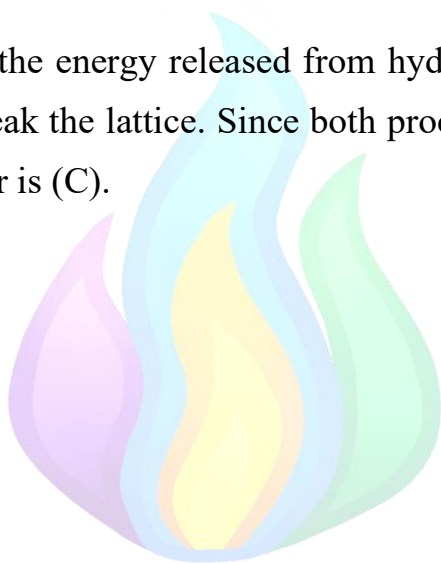
Question 25

When NaCl dissolves in water, two main energy changes occur:

1. Lattice Energy (Endothermic Process): Energy is absorbed to break the strong ionic bonds in the NaCl crystal lattice, separating Na^+ and Cl^- ions.
2. Hydration Energy (Exothermic Process): Energy is released when water molecules surround and stabilize the dissolved Na^+ and Cl^- ions through ion-dipole interactions.

For dissolution to occur, the energy released from hydration must compensate for the energy required to break the lattice. Since both processes result in a net energy release, the correct answer is (C).

Correct Answer: C



Question 26

Isotopes of an element have the same number of protons but different numbers of neutrons, resulting in different atomic masses. The atomic mass of an element with two isotopes is calculated using the weighted average formula:

$$\text{Atomic Mass} = \frac{M_1 \times P_1 + M_2 \times P_2}{100}$$

where:

M_1 and M_2 are the atomic masses of the isotopes

P_1 and P_2 are their respective mass percentages.

From the mass spectrum, we observe two peaks:

One at 20 amu with a mass percentage of 90%,

Another at 22 amu with a mass percentage of 10%.

Using the formula:

$$\text{Atomic Mass} = \frac{20 \times 90 + 22 \times 10}{100} = \frac{1800 + 220}{100} = \frac{2020}{100} = 20.2$$

Correct Answer: B

Question 27

To determine the mass of NaOH required to prepare 250 mL (or 0.250 L) of a 2 M solution, we use the formula:

$$\text{Moles} = \text{Concentration} \times \text{Volume} = 2 \text{ M} \times 0.250 \text{ L} = 0.5 \text{ moles}$$

Since the molar mass of NaOH is 40 g/mol, the required mass is:

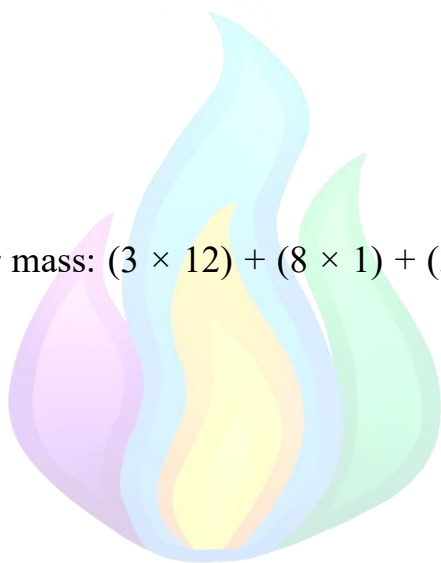
$$\text{Mass} = \text{Moles} \times \text{Molar Mass} = 0.5 \times 40 = 20 \text{ g}$$

Correct Answer: - A

Question 28

Glycerol ($\text{C}_3\text{H}_8\text{O}_3$) molar mass: $(3 \times 12) + (8 \times 1) + (3 \times 16) = 36 + 8 + 48 = 92$ g/mol

Correct Answer: A



Question 29

The reaction shows that 160 g of Fe_2O_3 produces 112 g of Fe. Using this ratio, we calculate how much Fe_2O_3 is needed for 28000g of Fe, which comes out to 40000g.

$$\text{Molar mass of } \text{Fe}_2\text{O}_3 = (2 \times 56) + (3 \times 16) = 160 \text{ g}$$

From the equation: $160 \text{ g } \text{Fe}_2\text{O}_3 \rightarrow 112 \text{ g Fe}$

For 28 kg (28,000 g) Fe: Required $\text{Fe}_2\text{O}_3 = (160/112) \times 28000 = 40000 \text{ g}$

Correct Answer: C

Question 30

Fructose is the sweetest naturally occurring sugar, while lactose is the least sweet.

Correct Answer: B

Question 31

$$W = F \times \text{distance}$$

$$F = \text{gravity} = 2\text{kg} \times 10\text{m/s}^2 = 20\text{N} \text{ Therefore, you are applying } 20\text{N}.$$

Since the force is constant and the directions line up,

$$(\text{Work done}) = F \times 1\text{m} = 20\text{N} \times 1\text{m} = 20\text{ J}$$

Correct Answer: D

Question 32

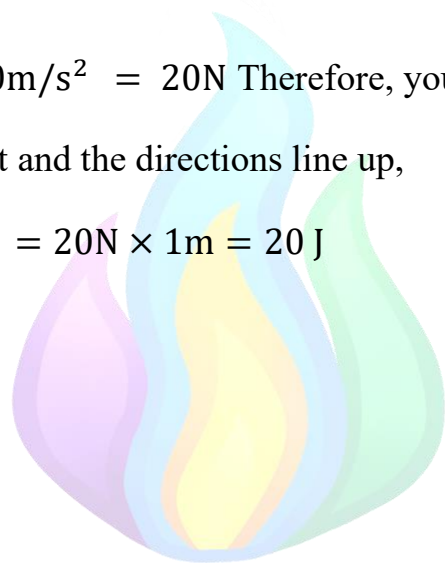
$$\text{distance} = \text{velocity} \times \text{time}$$

$$= 3.0 \times 10^8 \text{m/s} \times 2.0\text{ms} \times \frac{1\text{s}}{10^3 \text{ m/s}}$$

$$(3.0 \times 10^8 \text{m/s} \times 2.0 \times \frac{1}{10^3})\text{m} = 6 \times 10^5 \text{m}$$

$$6 \times 10^5 \text{m} \times \frac{1\text{km}}{10^3 \text{m}} = 600\text{km}$$

Correct Answer: D



Question 33

voltage = (resistance₁)(current₁) + (resistance₂)(current₂) + ...

The voltage between X, Y is the same as the voltage of (5Ω + 3Ω)

Since the current in the 5Ω is equal to the current in the 3Ω, if we call both I

$$4V = I \times 5\Omega + I \times 3\Omega$$

$$I = \frac{4V}{(5 + 3)\Omega} = 0.5V/\Omega = 0.5A$$

Correct Answer: C

Question 34

Solution 1:

$$v = \sqrt{\frac{\tau}{\mu}}$$

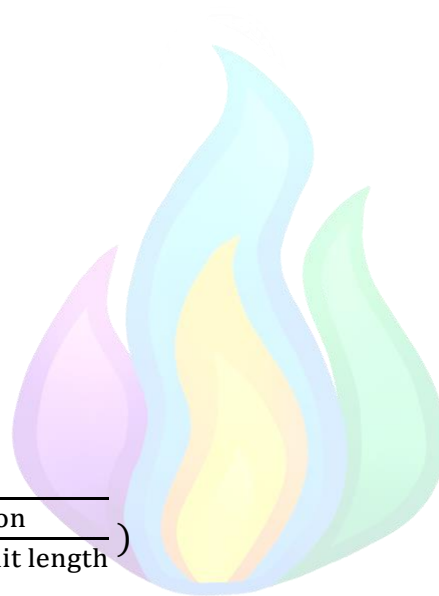
$$(\text{velocity} = \sqrt{\frac{\text{tension}}{\text{mass per unit length}}})$$

$$m = \mu = \frac{\text{mass}}{\text{length}}$$

(m = mass per unit length)

The unit of m is kg. m⁻¹

Correct Answer: C



Question 35

Only pressure changed.

Since temperature is assumed to be constant, we can use Boyle's Law:

$$(5\text{cm}^3) \times (6 \times 10^5\text{Pa}) = V \times (1.0 \times 10^5\text{Pa})$$

$$V = 30\text{cm}^3$$

Correct Answer: D

Question 36

$$Q = cm\Delta T$$

$$Q = 440\text{J/kg}^\circ\text{C} \times m\text{kg} \times (72^\circ\text{C} - 21^\circ\text{C}) : \text{for A}$$

$$160\text{J/kg}^\circ\text{C} \times m\text{kg} \times (x^\circ\text{C} - 21^\circ\text{C}) : \text{for B}$$

$$440m \times (72 - 21) = 160m \times (x - 21)$$

$$x = 161.25 \approx 160^\circ\text{C}$$

Correct Answer: A

Question 37

DELETED QUESTION

Question 38

Area under the curve is the distance travelled. By symmetry, you can see the answer should be 4m.

Correct Answer: D

Question 39

$$P = P_0 + \rho gh$$

In the same fluid, as x increases, h decreases.

And P decreases in a linear relation, slope being: $-\rho g$

Liquid M: slope is $-2\rho g$

Liquid L: slope is $-\rho g$; half of slope M

atmosphere: slope is $-\rho_{\text{gas}}g$; but ρ_{gas} is so small compared to the pressure at the surface, so the decrease is negligible.

However, no option fits.

The answer key provided by SLJSO claims C to be the answer, but a mistake was likely made in the memo.

Correct Answer:

Question 40

$$\text{Input heat} = 5\text{kW} \times 10\text{min} = 3 \times 10^6\text{J}$$

$$3 \times 10^6\text{J} = 1.2 \times 10^5\text{J} \cdot \text{K}^{-1} \cdot 20\text{K} + x\text{J} \cdot \text{K}^{-1} \times 20\text{K}$$

$$3 \times 10^6\text{J} = (1.2 \times 10^5 \times 20 + x \times 20)\text{J}$$

$$3 \times 10^6\text{J} = (20x + 2.4 \times 10^6)\text{J}$$

$$3 \times 10^6 = 20x + 2.4 \times 10^6$$

$$x = 3 \times 10^4$$

$$3 \times 10^4\text{J} \cdot \text{K}^{-1}$$

Correct Answer: D

Question 41

A convex mirror always produces an upright, diminished, virtual image. So, the correct answer is B.

Correct Answer: B



Question 42

When moving at constant velocity

$$P = Fv$$

$$2\text{kW} = x\text{N} \times 10\text{m/s} = 10x\text{Nm/s} = 10x\text{W}$$

$$2\text{kW} \times \frac{10^3\text{W}}{1\text{kW}} = 10x\text{W}$$

$$2 \times 10^3 = 10x$$

$$x = 200\text{N}$$

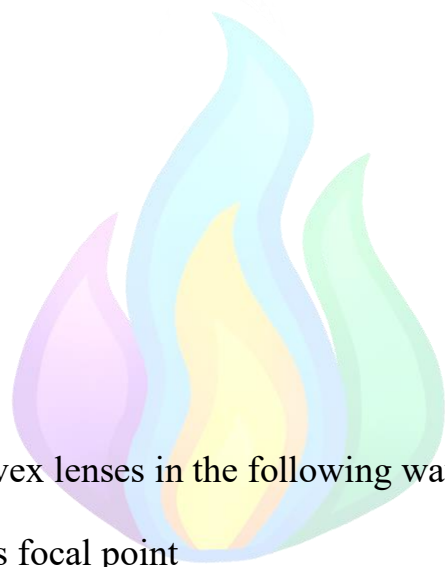
Correct Answer: C

Question 43

Light passes through convex lenses in the following way. If the ray

- Is Parallel $\rightarrow$ passes focal point
- Passes focal point $\rightarrow$ goes parallel
- Passes middle of lens $\rightarrow$ goes straight

Correct Answer: C



Question 44

By Newton's 2nd Law:

(net external force) = (total mass of system) $\times$ (acceleration of system)

$$20\text{N} = (1\text{kg} + 3\text{kg}) \times a \text{ m/s}^2$$

$a = 5$, so the system is accelerating at 5m/s^2

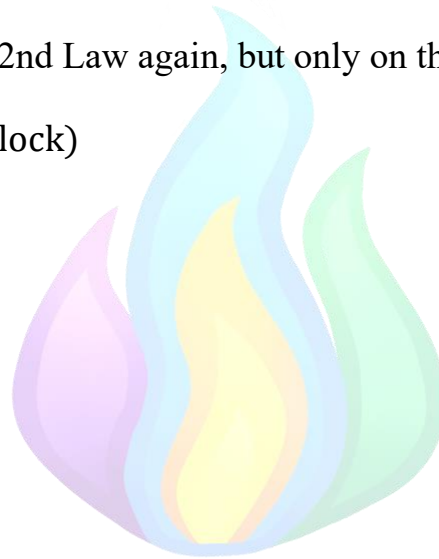
Here, each component is being accelerated at 5 m/s^2 as well.

So if we apply Newton's 2nd Law again, but only on the single block,

$$T = 1\text{kg} \times 5\text{m/s}^2 \text{ (left block)}$$

Therefore, $T = 5\text{N}$

Correct Answer: C



Question 45

(force by all legs) = (force by 100kg – mass)

$$= 100\text{kg} \times 10\text{m/s}^2 = 1000\text{kgm/s}^2 = 1000\text{N}$$

$$\text{(pressure by all legs)} = \frac{1000\text{N}}{20\text{cm}^2 \times \left(\frac{1\text{m}}{100\text{cm}}\right)^2} = 5 \times 10^5 \text{N/m}^2 = 5 \times 10^5 \text{Pa}$$

$$\text{pressure by 1 leg} = \frac{5 \times 10^5 \text{Pa}}{4} = 1.25 \times 10^5 \text{Pa}$$

Correct Answer: C

Question 46

$$\text{current} = \frac{\text{charge}}{\text{time}}$$

$$3.0\mu\text{A} \times \frac{1\text{A}}{10^6\mu\text{A}} = \frac{x\text{C}}{1.5 \times 60}$$

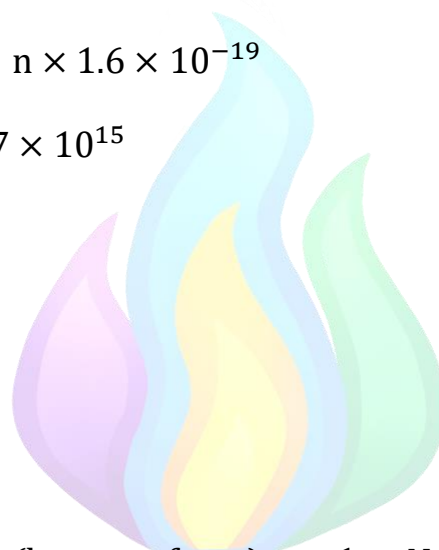
$$\frac{3}{10^6} = \frac{x}{1.5 \times 60}$$

$$x = 2.7 \times 10^{-4}\text{C}$$

$$\text{Therefore: } 2.7 \times 10^{-4} = n \times 1.6 \times 10^{-19}$$

$$n = 1.6875 \times 10^{15} \approx 1.7 \times 10^{15}$$

Correct Answer: B



Question 47

$x\text{N}$: water pushing stone (buoyant force) and $y\text{N}$: stone pushing water and
 $z\text{N}$: electronic balance pushing water (normal force)

Since net force of stone and water is 0,

$$5 = 2 + x \text{ (stone)}$$

$$10 + y = z \text{ (water)}$$

$$x = y \text{ (Newton's 3rd Law)}$$

Therefore: $x = y = 3$; $z = 13$. Therefore: normal force = 13N.

Correct Answer: B

Question 48

Conservation of energy: $\Delta(\text{kinetic energy}) + \Delta(\text{potential energy}) = 0$

$$\left(\frac{1}{2}mv^2 - \frac{1}{2}m \cdot 0^2\right) + (m \cdot g \cdot 0 - m \cdot g \cdot R) = 0$$

$$\frac{1}{2}mv^2 - mgR = 0$$

$$v = \sqrt{2gR}$$

Correct Answer: B

Question 49

DELETED QUESTION



Question 50

We have to rearrange this equation.

$$T = 2\pi \sqrt{\frac{l}{g + a}}$$

$$\frac{T}{2\pi} = \sqrt{\frac{l}{g + a}}$$

Square both sides,

$$\left(\frac{T}{2\pi}\right)^2 = \frac{l}{g + a}$$

Swap the numerator and denominator,

$$\left(\frac{2\pi}{T}\right)^2 = \frac{g + a}{l}$$

$$\left(\frac{2\pi}{T}\right)^2 l = g + a$$

$$\left(\frac{2\pi}{T}\right)^2 l - a = g$$



Correct Answer: B